Miracle Agbebaku

Software engineer: AI agents, backend and automation

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SUMMARY

Software engineer who builds AI-agent and messaging systems where the model proposes and deterministic code decides: structured plans, policy checks, audit trails and tests. Works in TypeScript and Python and on edge infrastructure. Three systems so far: an agent governance pipeline, a multi-account WhatsApp assistant and a social platform backend.

PROJECTS

NEXUS CORE

Working prototype

Next.js, React, TypeScript, FastAPI, Pydantic, SQLite

Governance-first AI agent orchestration: the model plans, a deterministic fail-closed policy engine decides.

- Designed the pipeline so no model-generated plan executes without passing a deterministic, fail-closed policy check. Blocked plans get at most two bounded corrections, each checked again.
- Streams every pipeline stage to the interface over Server-Sent Events and records SHA-256 hash-linked audit events in SQLite.
- Validates model output with Pydantic schemas. Execution is mostly simulated, with read-only web search as the one real tool, and the limits are documented.

ELLA

Active development

TypeScript, Node.js, Express, Supabase, React, Vite

WhatsApp assistant built around NEXUS CORE, running many linked accounts in one process.

- Isolated each linked account's session and scoped every chat, document and memory lookup by account and chat.
- Added a risk-tiered permission engine (five roles, four risk levels) and a response guard that strips internal data before every outgoing message.
- Built a provider gateway with ordered fallback and health tracking, Nigerian English and Pidgin handling, hostile-URL rules for link tools and about 50 test scripts.

Lagoon

In development

JavaScript, Cloudflare Workers, Durable Objects, WebSockets, Playwright

Social platform on a modular JavaScript frontend and a Cloudflare edge backend.

- Implemented live messaging over WebSockets with per-user Durable Objects, and client-side encryption for one-to-one chats using browser crypto (ECDH P-256, HKDF, AES-GCM). Groups are sealed per member.
- Kept moderated Spaces deliberately readable and labelled as not end-to-end encrypted, so each place makes a clear privacy promise.
- Built a registry-based frontend with 25 interchangeable UI shells, tested with browser suites across phone, tablet and desktop sizes.

SKILLS

Languages	Python, TypeScript, JavaScript, HTML5, CSS
Frontend	React, Next.js, Vite, Tailwind CSS, Three.js
Backend and APIs	FastAPI, Pydantic, Node.js, Express, Zod, Webhooks
Realtime and edge	Cloudflare Workers, Durable Objects, WebSockets, Server-sent events
Data and auth	PostgreSQL, SQLite, Supabase, Clerk
Testing and delivery	Playwright, Vitest, GitHub Actions, Git, Vercel, Render
Integrations	WhatsApp, FFmpeg
Practices	Policy gates, audit logging, input validation, streaming interfaces, automated testing

WRITING

20 engineering notes on trust boundaries, AI agent pipelines, messaging systems and how the portfolio is built, at miracle-agbebaku-portfolio.vercel.app/engineering.